

DAPP AGDM, AB, Canada

WMO: 713400

Lat: 54.3290N

Lon: 113.9345W

Elev: 619

StdP: 94.11

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-34.2	-30.5	-37.7	0.1	-34.2	-33.8	0.2	-30.4	10.6	-0.1	9.3	-1.7	1.7	100	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	27.5	17.8	25.8	17.1	24.3	16.3	19.9	25.5	18.6	24.0	17.4	22.6	3.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.7	23.2	16.4	12.6	21.7	15.2	11.6	20.0	59.7	25.5	55.2	23.9	51.3	22.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.4	7.3		-40.2	30.9	2.9	1.6	-42.3	32.0	-44.0	32.9	-45.7	33.8	-47.8	35.0
				-40.3	22.3	2.9	1.5	-42.4	23.3	-44.1	24.2	-45.7	25.1	-47.9	26.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-12.8	-12.4	-6.6	2.7	9.9	13.9	15.8	14.2	9.4	2.9	-6.1	-12.9
	DBStd	12.23	8.50	7.92	7.49	5.49	4.39	2.93	2.73	3.13	4.18	4.61	7.19	7.66
	HDD10.0	3611	706	626	516	221	56	4	1	4	60	224	484	710
	HDD18.3	6132	964	859	774	468	262	137	87	132	267	479	734	968
	CDD10.0	537	0	0	0	3	53	120	181	134	43	3	0	0
	CDD18.3	16	0	0	0	0	1	3	9	3	0	0	0	0
	CDH23.3	546	0	0	0	2	57	98	208	131	48	2	0	0
	CDH26.7	92	0	0	0	0	9	14	40	20	8	0	0	0
(14) Wind	WSAvg	3.2	3.0	3.0	3.4	3.9	4.0	3.5	2.8	2.7	3.1	3.3	3.1	2.8
(15) Precipitation	PrecAvg	431	22	11	17	25	49	82	82	56	35	19	20	18
	PrecMax	538	93	23	38	49	79	134	161	118	79	34	70	33
	PrecMin	321	0	0	3	1	7	32	38	22	4	2	3	1
	PrecStd	71	22	7	10	13	19	29	31	22	19	10	17	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	5.6	9.5	22.5	27.9	28.5	30.2	28.9	27.8	21.8	11.3	5.9
		MCWB	2.9	2.4	5.0	11.0	13.5	16.7	20.2	18.4	16.4	12.1	6.5	2.8
	2%	DB	4.2	3.5	6.5	17.9	24.8	26.0	27.5	26.6	24.4	16.3	7.1	3.1
		MCWB	1.6	0.8	3.0	8.8	12.6	16.1	19.7	18.2	15.0	9.3	3.7	0.8
	5%	DB	2.2	1.4	4.6	14.6	22.2	24.0	25.5	24.6	21.6	13.3	4.7	0.9
		MCWB	0.3	-0.5	1.9	7.2	11.3	15.0	18.6	17.4	13.7	7.5	2.0	-0.9
	10%	DB	-0.4	-0.6	2.9	11.9	19.9	21.9	23.9	22.7	18.5	10.8	2.5	-1.6
		MCWB	-1.8	-2.1	0.7	6.0	10.4	14.1	17.6	16.4	11.9	6.2	0.8	-2.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.7	5.7	11.5	14.8	19.1	22.2	20.8	17.3	12.4	6.8	2.9
		MCDB	5.7	5.3	8.9	21.0	24.9	24.8	27.4	25.8	25.7	21.0	10.6	5.5
	2%	WB	1.8	1.1	3.2	9.2	13.3	17.2	20.6	19.3	15.6	9.8	4.1	0.9
		MCDB	4.0	3.1	6.0	16.7	22.3	23.6	26.1	25.0	23.0	14.9	6.9	3.0
	5%	WB	0.4	-0.3	2.0	7.6	12.2	16.0	19.3	18.0	14.0	8.1	2.3	-0.9
		MCDB	2.2	1.5	4.3	13.8	20.5	21.8	24.6	23.3	20.7	12.3	4.4	0.7
	10%	WB	-1.8	-2.2	0.9	6.1	11.1	14.9	18.1	16.9	12.6	6.7	0.9	-2.8
		MCDB	-0.4	-0.6	2.6	11.1	18.5	20.4	23.0	22.0	17.5	10.3	2.3	-1.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.5	11.8	11.9	14.5	13.2	13.8	14.6	14.8	12.4	9.8	10.5
		MCDBR	12.1	14.0	12.7	17.8	19.6	17.6	17.4	18.3	20.7	17.5	12.2	12.5
		MCWBR	10.0	11.8	9.6	9.8	9.3	8.8	10.4	10.7	11.5	10.8	8.8	10.7
	5% WB	MCDBR	12.0	13.3	12.1	16.5	17.3	14.8	16.3	16.5	19.2	16.5	11.0	12.4
		MCWBR	10.1	11.3	9.3	9.5	8.5	8.2	10.2	10.1	10.8	10.5	8.4	10.7
(40) Clear-Sky Solar Irradiance	taub		0.253	0.274	0.305	0.327	0.343	0.355	0.357	0.361	0.310	0.283	0.253	0.236
	taud		2.285	2.297	2.313	2.391	2.399	2.376	2.421	2.355	2.510	2.557	2.453	2.294
	Ebn at Noon		719	827	876	903	903	892	883	855	869	824	735	666
	Edh at Noon		56	78	97	104	109	113	106	107	80	61	47	44
(44) All-Sky Solar Radiation	RadAvg		0.86	1.87	3.35	4.63	5.61	5.88	5.94	4.90	3.39	1.91	0.90	0.62
	RadStd		0.09	0.14	0.26	0.39	0.46	0.41	0.34	0.36	0.35	0.23	0.09	0.05

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page